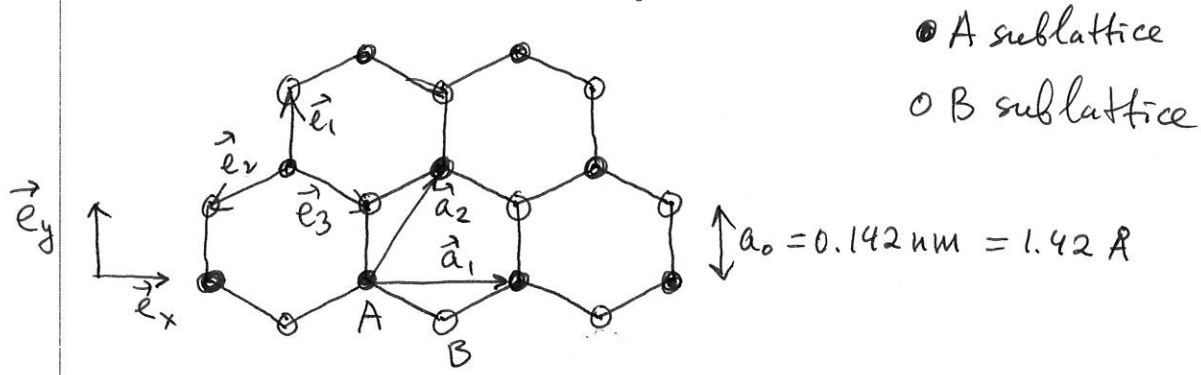
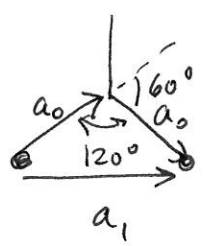
Graphene structure: honeycomb lattice



Honeycomb is not a Bravais lattice: sites A and B however form triangular Bravais lattice with basis vectors $\vec{a}_1, \vec{a}_2$.
Any site on A lattice: $\vec{R}_A = n_1 \vec{a}_1 + n_2 \vec{a}_2$.



$$|a_1|^2 = |a_0|^2 + |a_0|^2 + 2|a_0|^2 \underbrace{\cos\frac{2\pi}{3}}_{+1/2} = 3|a_0|^2$$

$$\Rightarrow a_2 = a_1 = \sqrt{3}\, a_0$$

Reciprocal lattice formed by $\vec{b}_1, \vec{b}_2$ such that

$$\vec{a}_i \cdot \vec{b}_j = 2\pi\, \delta_{ij}$$

Any vector $\vec{K}$ of reciprocal lattice $\vec{K} = m_1 \vec{b}_1 + m_2 \vec{b}_2$.
Hence $\vec{K} \cdot \vec{R}_A = (m_1 \vec{b}_1 + m_2 \vec{b}_2) \cdot (n_1 \vec{a}_1 + n_2 \vec{a}_2) =$

$$= m_1 n_1 \underbrace{\vec{b}_1 \cdot \vec{a}_1}_{2\pi} + m_1 n_2 \underbrace{\vec{b}_1 \cdot \vec{a}_2}_{0} + m_2 n_1 \underbrace{\vec{b}_2 \cdot \vec{a}_1}_{0}$$

$$+ m_2 n_2 \underbrace{\vec{b}_2 \cdot \vec{a}_2}_{2\pi} = 2\pi (m_1 n_1 + m_2 n_2)$$

$$\Rightarrow e^{i \vec{K} \cdot \vec{R}_A} = e^{i 2\pi (m_1 n_1 + m_2 n_2)} = 1.$$

A and B sites are connected by $(\vec{e}_1, \vec{e}_2, \vec{e}_3)$; $|\vec{e}_i| = e_i = a_0$

In terms of orthogonal basis $\vec{e}_x, \vec{e}_y$: $\vec{e}_1 = +a_0 \vec{e}_y$

$$\vec{e}_2 = -\frac{a_0}{2}\left(\sqrt{3}\,\vec{e}_x + \vec{e}_y\right)$$

$$\vec{e}_3 = \frac{a_0}{2}\left(\sqrt{3}\,\vec{e}_x - \vec{e}_y\right)$$

Triangular basis vectors are

$$\vec{a}_1 = \sqrt{3}\,a_0\,\vec{e}_x\,, \quad \vec{a}_2 = \frac{\sqrt{3}\,a_0}{2}\left(\vec{e}_x + \sqrt{3}\,\vec{e}_y\right)$$

so that $|\vec{a}_{1,2}| = \sqrt{3}\,a_0$.

General construction for $\vec{b}_i$:

$$\vec{b}_1 = 2\pi \frac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

$$\vec{b}_2 = 2\pi \frac{\vec{a}_3 \times \vec{a}_1}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

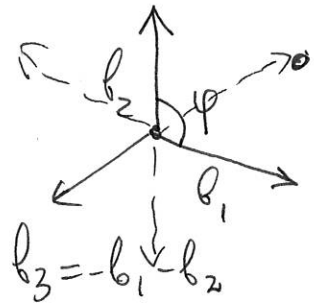
$$\vec{b}_3 = 2\pi \frac{\vec{a}_1 \times \vec{a}_2}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

Add $\vec{a}_3 = a_0 \vec{e}_z = a_0(0, 0, 1)$ so that triangular lattice repeats along z-axis

Volume $v = \vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3 = a_0^3 \begin{pmatrix} \sqrt{3} & 0 & 0 \\ \frac{\sqrt{3}}{2} & \frac{3}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \frac{3\sqrt{3}}{2} a_0^3$

$$\vec{b}_1 = \frac{2\pi}{\frac{3\sqrt{3}}{2} a_0^3} \cdot \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\sqrt{3}}{2} & \frac{3}{2} & 0 \\ 0 & 0 & 1 \end{vmatrix} a_0^2 = \frac{2\pi}{\sqrt{3}\,a_0}\left(\vec{e}_x - \frac{1}{\sqrt{3}}\vec{e}_y\right)$$

$$\vec{b}_2 = \frac{2\pi}{\frac{3\sqrt{3}}{2} a_0^3} \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ 0 & 0 & 1 \\ \sqrt{3} & 0 & 0 \end{vmatrix} a_0^2 = \frac{4\pi}{3a_0}\vec{e}_y$$



$$|\vec{b}_1| = \frac{2\pi}{\sqrt{3}a_0}\left(1+\frac{1}{3}\right)^{1/2} = \frac{2\pi}{\sqrt{3}a_0}\sqrt{\frac{4}{3}} = \frac{4\pi}{3a_0}$$

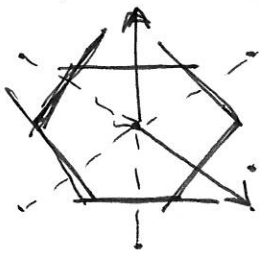
$$|\vec{b}_2| = \frac{4\pi}{3a_0}$$

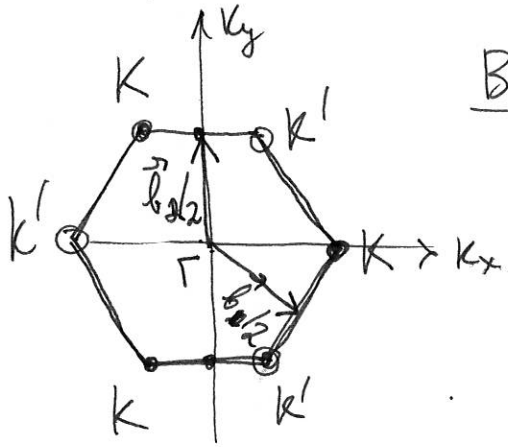
$$\vec{b}_1 \cdot \vec{b}_2 = \left(-\frac{2\pi}{3a_0}\vec{e}_y\right)\cdot\left(\frac{4\pi}{3a_0}\vec{e}_y\right) = +\left(\frac{4\pi}{3a_0}\right)^2 \underbrace{\cos\varphi}_{-1/2}$$

$\Rightarrow$ angle between $\vec{b}_1$ and $\vec{b}_2$ is $\frac{2\pi}{3}$.

Reciprocal lattice ~~is~~ triangular lattice too.

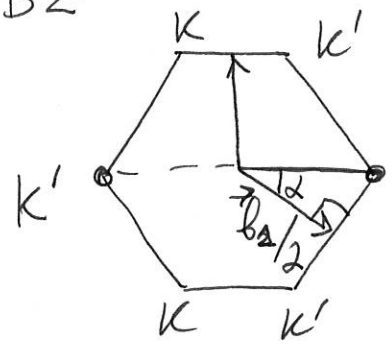
$\Rightarrow$ build reciprocal triang. lattice and then BZ:





Brillouin Zone

Corners of BZ



$$\vec{K} = \frac{4\pi\,\hat{x}}{3\sqrt{3}a_0} = \frac{4\pi}{3a}\,\hat{x}$$

$$\frac{1}{2}\vec{b}_1 \cdot \vec{e}_x = \frac{1}{2}\frac{2\pi}{\sqrt{3}a_0} = \frac{2\pi}{\sqrt{3}a_0}\cdot\frac{1}{2} \Rightarrow \cos\alpha = \frac{1}{2} \Rightarrow \alpha = \pi/3$$

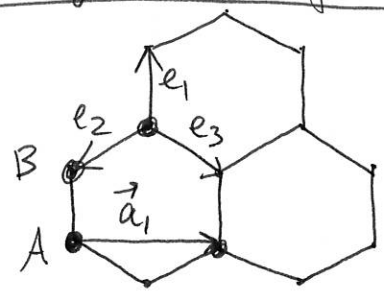
But $K \sin\alpha = \frac{b_1}{2} = \frac{2\pi}{3a_0}$

$$K \sin\frac{\pi}{3} = \frac{2\pi}{3a_0}, \quad K = \frac{4\pi}{3\sqrt{3}a_0}$$

$$\Rightarrow \vec{K} = \frac{4\pi}{3\sqrt{3}a_0}\vec{e}_x, \quad \vec{K}' = -\frac{4\pi}{3\sqrt{3}a_0}\vec{e}_x$$

All others can be obtained by reciprocal lattice translations.

Tight-binding model for graphene



In terms of $|\vec{a}_1| = \sqrt{3}\, a_0 \equiv a$, vectors connecting A and B sublattices are $(a_0 = a/\sqrt{3})$

$$\vec{e}_1 = \frac{a}{\sqrt{3}}\,\vec{e}_y = \frac{a}{\sqrt{3}}\,\hat{y}$$

$$\vec{e}_2 = -\frac{a}{2\sqrt{3}}\left(\sqrt{3}\,\vec{e}_x + \vec{e}_y\right) = a\left(-\frac{1}{2}\hat{x} - \frac{1}{2\sqrt{3}}\hat{y}\right)$$

$$\vec{e}_3 = \frac{a}{2\sqrt{3}}\left(\sqrt{3}\,\vec{e}_x - \vec{e}_y\right) = a\left(\frac{1}{2}\hat{x} - \frac{1}{2\sqrt{3}}\hat{y}\right)$$

Let $c^{+}_{A/B,\vec{r},\sigma}$ create an electron with spin σ on the carbon p_z orbital on site $\vec{r}$ of A/B sublattice

$$H = -t \sum_{\langle rr' \rangle} \sum_{\sigma} \left(c^{+}_{A\vec{r}\sigma}\, c_{B\vec{r}'\sigma} + c^{+}_{B\vec{r}'\sigma}\, c_{A\vec{r}\sigma} \right)$$

sum over $\langle \vec{r}\,\vec{r}' \rangle$ = neighboring pairs of A-B sites

t = tunneling matrix element $= 2.97\,\text{eV}$
(determined by overlap between neighboring p_z orbitals)

$$H = -t \sum_{\sigma} \sum_{\vec{r} \in A} \sum_{j=1}^{3} \left(c^{+}_{A\vec{r}\sigma}\, c_{B\,\vec{r}+\vec{e}_i\,\sigma} + h.c. \right)$$

Transform to $\vec{k}$ space

$$c_{\alpha \vec{r} \sigma} = \frac{1}{\sqrt{N_A}} \sum_{\vec{k} \in BZ} e^{i \vec{k} \cdot \vec{r}} c_{\alpha \vec{k} \sigma}$$

$\alpha = A$ or B ; N_A = number of unit cells = $\frac{1}{2}$ Number of sites.

$$H = -t \sum_{\sigma} \sum_{k, k'} \sum_{j=1}^{3} \frac{1}{N_A} \sum_{\vec{r} \in A} \Big(e^{-i \vec{k} \cdot \vec{r}} e^{i \vec{k}' (\vec{r} + \vec{e}_j)} c^{+}_{A \vec{k} \sigma} \cdot c_{B \vec{k}' \sigma} + h.c. \Big)$$

$$\frac{1}{N_A} \sum_{\vec{r} \in A} e^{i \vec{r} \cdot (\vec{k}' - \vec{k})} = \delta_{\vec{k}, \vec{k}'}$$ (upto reciprocal lattice vector $\vec{G}$ which takes one of $\vec{k}$'s outside the BZ)

$$\Rightarrow H = -t \sum_{\vec{k} \in BZ} \sum_{j=1}^{3} \sum_{\sigma} \cdot \Big(e^{i \vec{k} \cdot \vec{e}_j} c^{+}_{A k \sigma} c_{B k \sigma} + e^{-i \vec{k} \cdot \vec{e}_j} c^{+}_{B k \sigma} c_{A k \sigma} \Big)$$

Note $$\sum_{j=1}^{3} e^{i \vec{k} \cdot \vec{e}_j} = e^{i \vec{k} \cdot \vec{e}_1} + e^{i \vec{k} \cdot \vec{e}_2} + e^{i \vec{k} \cdot \vec{e}_3} =$$

$$= e^{i k_y a / \sqrt{3}} + e^{-i k_x a/2 - i k_y a / 2\sqrt{3}} + e^{i k_x a/2 - i k_y a / 2\sqrt{3}}$$

$$= 2 \cos\left(\frac{k_x a}{2}\right) e^{-i \frac{k_y a}{2\sqrt{3}}} + e^{i \frac{k_y a}{\sqrt{3}}}$$

$$H = \sum_{\vec{k} \in BZ} \sum_{\sigma} \Big(\xi(\vec{k}) c^{+}_{A \vec{k} \sigma} c_{B \vec{k} \sigma} + \xi^{*}(\vec{k}) c^{+}_{B \vec{k} \sigma} c_{A \vec{k} \sigma} \Big)$$

$$\xi(\vec{k}) \equiv -t \sum_{j=1}^{3} e^{i \vec{k} \cdot \vec{e}_j}$$

Define $c_\sigma(\vec{k}) = \begin{pmatrix} c_{A\vec{k}\sigma} \\ c_{B\vec{k}\sigma} \end{pmatrix}$

$$c^\dagger_\sigma(\vec{k}) = \left(c^\dagger_{A\vec{k}\sigma},\ c^\dagger_{B\vec{k}\sigma}\right)$$

$$H = \sum_{\vec{k},\sigma} c^\dagger_\sigma(\vec{k}) \underbrace{\begin{pmatrix} 0 & \xi(k) \\ \xi^*(k) & 0 \end{pmatrix}}_{\mathcal{H}(\vec{k})} c_\sigma(k)$$

Eigenvalues of $\mathcal{H}(\vec{k})$

$$\det\left(\mathcal{H}(\vec{k}) - E(\vec{k})I\right) = 0 = \det\begin{pmatrix} -E & \xi(k) \\ \xi^*(k) & -E \end{pmatrix}$$

$$\Rightarrow E^2 - |\xi(k)|^2 = 0$$

$$E_\pm(\vec{k}) = \pm|\xi(\vec{k})| =$$

$$= \pm t\left| 2\cos\frac{k_x a}{2}\, e^{-i\frac{k_y a}{2\sqrt{3}}} + e^{i\frac{k_y a}{\sqrt{3}}} \right|$$

$$= \pm t\left(2\cos\frac{k_x a}{2}\, e^{-i\frac{k_y a}{2\sqrt{3}}} + e^{i\frac{k_y a}{\sqrt{3}}}\right)^{1/2} \cdot \left(2\cos\frac{k_x a}{2}\, e^{i\frac{k_y a}{2\sqrt{3}}} + e^{-i\frac{k_y a}{\sqrt{3}}}\right)^{1/2}$$

$$= \pm t\left[\left(2\cos\frac{k_x a}{2}\right)^2 + 1 + 2\cos\frac{k_x a}{2}\, e^{-i\frac{\sqrt{3}k_y a}{2}} + e^{i\frac{\sqrt{3}k_y a}{2}}\right]^{1/2}$$

$$\frac{1}{2\sqrt{3}} + \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}\,\frac{3}{2} = \frac{\sqrt{3}}{2}$$

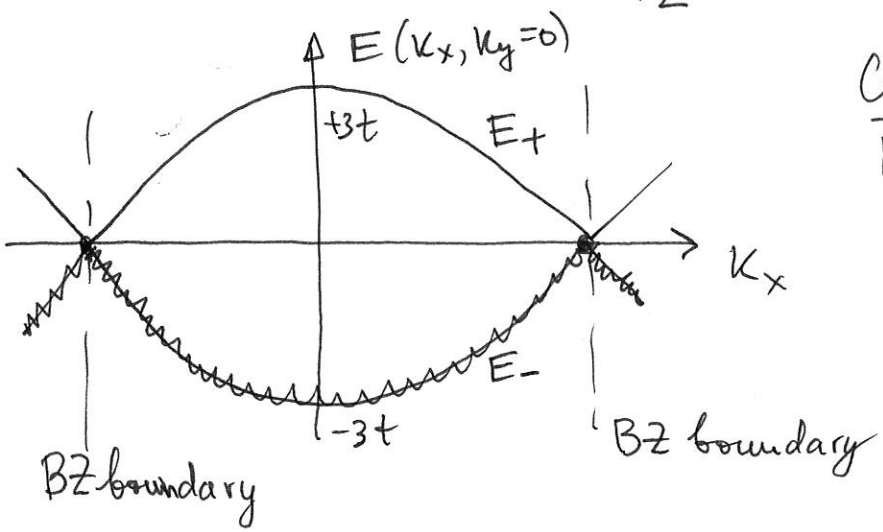
$$\underbrace{1+4\cos^2\frac{k_x a}{2}}_{2\left(1+\cos(k_x a)\right)} + 4\cos\frac{k_x a}{2}\cos\frac{\sqrt{3}}{2}k_y a = 3+2\cos(k_x a) + 4\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{\sqrt{3}}{2}k_y a\right)$$

$$\Rightarrow E_\pm(\vec{k}) = \pm t\sqrt{3+2\cos(k_x a)+4\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{\sqrt{3}}{2}k_y a\right)}$$

<u>Note</u>: 2 bands, one $E_+ \geq 0$, another $E_- \leq 0$
can touch only at points where $E=0$

By inspection: at $\vec{k} = \left(\pm\frac{4\pi}{3a}, 0\right) = \vec{K}$ or $\vec{K}'$

$$3+2\underbrace{\cos\left(\frac{4\pi}{3}\right)}_{-1/2} + 4\underbrace{\cos\left(\frac{1}{2}\,\frac{4\pi}{3}\right)}_{-1/2} = 3-1-2=0$$



Cut through $\vec{k}_y = 0$

Every C atom contributes 1 electron (out of 2 possible states)
$\Rightarrow$ the band structure should be half-filled
$\Rightarrow \begin{cases} E_-(\vec{k}) \text{ band is filled} \\ E_+(\vec{k}) \text{ is empty} \end{cases}$

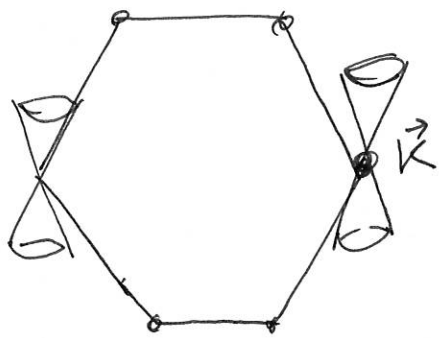
Near $\vec{K}$ ($\vec{K}'$) point: $\vec{k} = \vec{K} + \vec{q}$, $qa \ll 1$

$$E_\pm(\vec{K}+\vec{q}) = \pm t\sqrt{3 \cancel{+}2\cos\left(\frac{4\pi}{3} + q_x a\right) + 4\cos\left(\frac{2\pi}{3} + \frac{q_x a}{2}\right)\cdot\cos\frac{\sqrt{3}q_y a}{2}}$$

$$= \pm t\,\frac{\sqrt{3}}{2}\sqrt{(q_x a)^2 + (q_y a)^2}$$

$$= \pm \hbar v |\vec{q}| + O(\vec{q}^{\,2})$$ (relativistic Dirac spectrum)

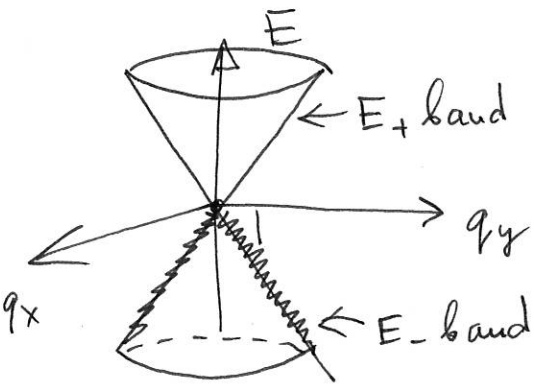
where $v = \frac{\sqrt{3}at}{2\hbar} \approx 10^6\ m/s \ll c$



"Dirac cones" at $\vec{K}, \vec{K}'$ points of BZ

c.f. Special relativity $E = \sqrt{m^2c^4 + p^2c^2} \xrightarrow[m=0]{} cp$

$\Rightarrow$ electrons in graphene: $c \to v \approx \frac{c}{300}$

$(3\cdot 10^8\ m/s) \to (10^6\ m/s)$

Near $\vec{K}$ point: